

Introduction to Biophysics II: Quantum physics & biology

Lecture 07: Energy transfer in photosynthesis - II

Förster Theory:

- Born-Oppenheimer approximation
- Nuclear interaction term: vibrations
- Electronic interaction term
- Transition density cube method
- Dipole-dipole interaction
- FRET rate

Reading:

Modern Optical Spectroscopy, chapter 7

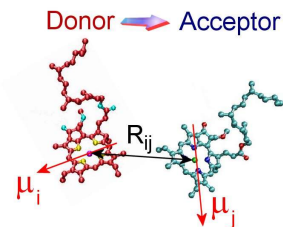
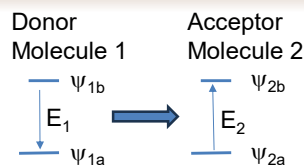
Additional:

Quantum Effects in Biology, chapter 3

1

Energy transfer between molecules.

Last time...



Förster Resonance Energy Transfer: FRET (1946)

Approach: an excited state of complex $(AB)^*$ can be described as a super-molecule with fixed nuclear positions and two valence electrons moving in supramolecular potential.

$$\text{Molecule 1 excited: } \Psi_1 = \Psi_{1b} \Psi_{2a}$$

$$\text{Molecule 2 excited: } \Psi_2 = \Psi_{1a} \Psi_{2b}$$

The initial and final states are delocalized over both molecules:

$$\Psi(t) = c_1 \Psi_1(t) + c_2 \Psi_2(t)$$

sometimes (biologists) call it fluorescence resonance energy transfer (FRET)
– not historically nor scientifically correct, there is no fluorescence involved.



Theodor Förster
1910 - 1974

2

FRET: spectral overlap

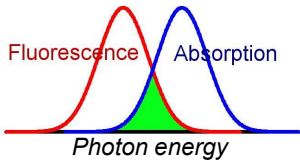
Last time...

$$k_{rt} \approx \frac{2\pi}{\hbar} \int_{-\infty}^{\infty} |\tilde{H}_{21}|^2 \rho_{s1}(E) \rho_{s2}(E) dE$$

For constant H_{21} can rewrite:

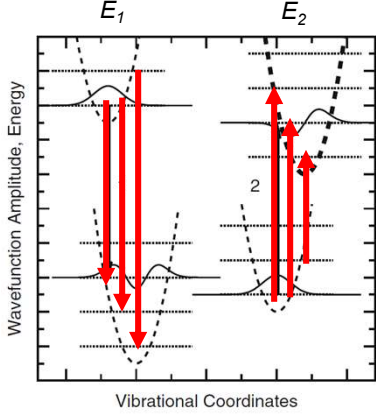
$$k_{rt} \approx \frac{2\pi}{\hbar} |\tilde{H}_{21}|^2 \int_{-\infty}^{\infty} \rho_{s1}(E) \rho_{s2}(E) dE$$

Spectral overlap integral



Photon energy

Absorption spectrum: probes density of states in *acceptor*
 Fluorescence: probes density of states in *donor*



3

FRET: Born-Oppenheimer approximation

$$k_{rt} = \frac{2\pi}{\hbar} \int_{-\infty}^{\infty} |H_{21}|^2 \rho_{s2}(E_1) \rho_{s1}(E_1) dE_1$$

Note: H_{21} depends on both electronic and nuclear coordinates.

$$H_{21} = \langle \phi_{1a} \chi_{1a} \phi_{2b} \chi_{2b} | \tilde{H}_{21} | \phi_{1b} \chi_{1b} \phi_{2a} \chi_{2a} \rangle$$

Born-Oppenheimer approximation: the nuclei will not move significantly during the instant when the excitation energy jumps between the molecules.

→ H_{21} thus can be approximated as a product of a purely electronic interaction matrix element ($H_{21(e)}$) and two nuclear overlap integrals

$$H_{21} = \langle \phi_{1a} \phi_{2b} | \tilde{H}_{21} | \phi_{1b} \phi_{2a} \rangle \langle \chi_{1a} | \chi_{1b} \rangle \langle \chi_{2b} | \chi_{2a} \rangle$$

$$= H_{21(e)} \langle \chi_{1a} | \chi_{1b} \rangle \langle \chi_{2b} | \chi_{2a} \rangle$$

Note: densities of states are defined by $\langle \chi_{1a} | \chi_{1b} \rangle$ and $\langle \chi_{2b} | \chi_{2a} \rangle$

4

FRET: nuclear term

$$k_{rt} = \frac{2\pi}{\hbar} \int_{-\infty}^{\infty} |H_{21}|^2 \rho_{s2}(E_1) \rho_{s1}(E_1) dE_1$$

$$H_{21} = \langle \phi_{1a} \phi_{2b} | \tilde{H}_{21} | \phi_{1b} \phi_{2a} \rangle \langle \chi_{1a} | \chi_{1b} \rangle \langle \chi_{2b} | \chi_{2a} \rangle$$

$$= H_{21(el)} \langle \chi_{1a} | \chi_{1b} \rangle \langle \chi_{2b} | \chi_{2a} \rangle.$$

Transition strength of the donor from the *excited* vibrational state n to the ground vibrational state m :

$$|\langle \chi_{1m} | \chi_{1n} \rangle|^2$$

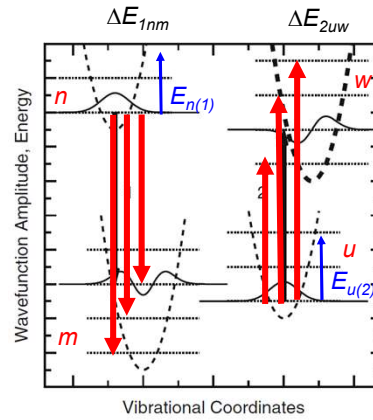
Probability to be populated at a certain temperature T :

$$\frac{\exp(-E_{n(1)}/k_B T)}{Z_1}$$

Need to account for all possible combinations of $m \rightarrow n$, each with transition energy ΔE_{1nm} :

$$\frac{\exp(-E_{n(1)}/kT)}{Z_1} |\langle \chi_{1m} | \chi_{1n} \rangle|^2$$

$$Z_1 = \sum_n \exp(-E_{n(1)}/kT) \quad \text{vibrational partition function (normalization, sum of all Boltzmann factors)}$$



5

FRET: nuclear term

$$k_{rt} = \frac{2\pi}{\hbar} \int_{-\infty}^{\infty} |H_{21}|^2 \rho_{s2}(E_1) \rho_{s1}(E_1) dE_1$$

$$H_{21} = \langle \phi_{1a} \phi_{2b} | \tilde{H}_{21} | \phi_{1b} \phi_{2a} \rangle \langle \chi_{1a} | \chi_{1b} \rangle \langle \chi_{2b} | \chi_{2a} \rangle$$

$$= H_{21(el)} \langle \chi_{1a} | \chi_{1b} \rangle \langle \chi_{2b} | \chi_{2a} \rangle.$$

For **donor**, transition energy ΔE_{1nm} :

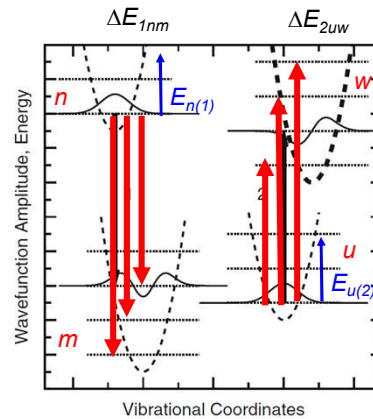
$$\frac{\exp(-E_{n(1)}/kT)}{Z_1} |\langle \chi_{1m} | \chi_{1n} \rangle|^2$$

For **acceptor**, transition energy ΔE_{2uw} :

$$\frac{\exp(-E_{u(2)}/kT)}{Z_2} |\langle \chi_{2w} | \chi_{2u} \rangle|^2$$

Combine all together and sum over *all* possible transition combinations:

$$k_{rt} = \frac{2\pi}{\hbar} |H_{21(el)}|^2 \sum_{n,m,u,w} \left[\frac{\exp(-E_{n(1)}/kT)}{Z_1} |\langle \chi_{1m} | \chi_{1n} \rangle|^2 \frac{\exp(-E_{u(2)}/kT)}{Z_2} |\langle \chi_{2w} | \chi_{2u} \rangle|^2 \delta(\Delta E_{1nm} - \Delta E_{2uw}) \right]$$



6

FRET: nuclear term

Born-Oppenheimer approximation:

$$H_{21} = \langle \phi_{1a} \phi_{2b} | \hat{H}_{21} | \phi_{1b} \phi_{2a} \rangle \langle \chi_{1a} | \chi_{1b} \rangle \langle \chi_{2b} | \chi_{2a} \rangle \\ = H_{21(ef)} \langle \chi_{1a} | \chi_{1b} \rangle \langle \chi_{2b} | \chi_{2a} \rangle.$$

$$k_{rt} = \frac{2\pi}{\hbar} |H_{21(ef)}|^2 \sum_{n,m,u,w} \left[\frac{\exp(-E_{n(1)}/kT)}{Z_1} |\langle \chi_{1m} | \chi_{1n} \rangle|^2 \times \frac{\exp(-E_{u(2)}/kT)}{Z_2} |\langle \chi_{2w} | \chi_{2u} \rangle|^2 \delta(\Delta E_{1nm} - \Delta E_{2uw}) \right]$$

Probability to find molecule 2 in **ground** state χ_u

Probability of transfer in molecule 2 from u of the ground state to w of the excited state

Probability of transfer in molecule 1 from n of the excited state to m of the ground state

Probability to find molecule 1 in **excited** state state χ_n

Energy conservation

The double sum with energy conservation term is the overlap integral (overlap in density of states)

$$k_{rt} \approx \frac{2\pi}{\hbar} |\tilde{H}_{21}|^2 \int_{-\infty}^{\infty} \rho_{s1}(E) \rho_{s2}(E) dE$$

Note: k_{rt} units are $[k_{rt}] = 1/s$

Keep in mind that:
 $[\rho] = 1/\text{Joule}$ (states per energy interval)
 $[dE] = \text{Joule}$
 $[\delta(E)] = 1/\text{Joule}$, since $\int_{-\infty}^{\infty} \delta(E) dE \equiv 1$

7

FRET: nuclear term

$$k_{rt} = \frac{2\pi}{\hbar} |H_{21(ef)}|^2 \sum_{n,m,u,w} \left[\frac{\exp(-E_{n(1)}/kT)}{Z_1} |\langle \chi_{1m} | \chi_{1n} \rangle|^2 \times \frac{\exp(-E_{u(2)}/kT)}{Z_2} |\langle \chi_{2w} | \chi_{2u} \rangle|^2 \times \delta(\Delta E_{1nm} - \Delta E_{2uw}) \right]$$

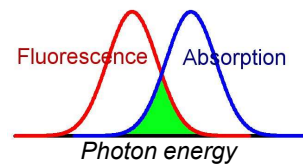
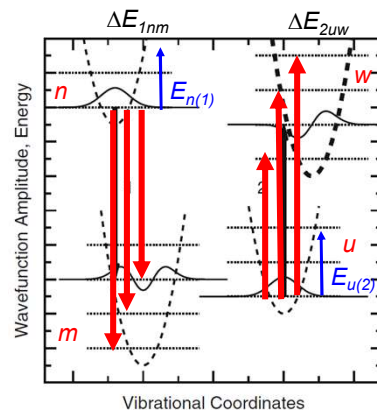
Note: the 1st term:

$$\frac{\exp(-E_{n(1)}/kT)}{Z_1} |\langle \chi_{1m} | \chi_{1n} \rangle|^2$$

Is the same as the fluorescence intensity of molecule 1 at each wavelength $\Delta E_{1nm} \rightarrow$ it is a fluorescence spectrum.

2nd term: absorption spectrum of molecule 2

The $\delta()$ means that the product of fluorescence intensity and absorption matters only at exact same energies. Thus, the sum represents the **overlap integral** of fluorescence and absorption spectra!



8

FRET: electronic interaction

$$k_{et} = \frac{2\pi}{\hbar} |H_{21(el)}|^2 \sum_{n,m,u,w} \left[\frac{\exp(-E_{n(l)}/kT)}{Z_1} |\langle \chi_{1m} | \chi_{1n} \rangle|^2 \times \frac{\exp(-E_{u(2)}/kT)}{Z_2} |\langle \chi_{2w} | \chi_{2u} \rangle|^2 \times \delta(E_{nm} - E_{uw}) \right]$$

Electronic interaction energy $H_{21(el)}$:

Assumption: no overlap between electronic wave functions of two molecules

→ Electrons can be assigned explicitly to molecule 1 or 2. I.e.:

There is no intermolecular exchange of electrons

The motion of electrons is not correlated

→ Dominant electronic interaction is coulombic.

Can estimate interaction as follows:

$$H_{21(el)} = \langle \phi_{1a} \phi_{2b} | \tilde{H}_{21} | \phi_{1b} \phi_{2a} \rangle \approx (f^2/n^2) \int \int \phi_{1a}^* \phi_{2b}^* \frac{e^2}{r_{21}} \phi_{1b} \phi_{2a} d\mathbf{r}_1 d\mathbf{r}_2$$

$\mathbf{r}_1$ and $\mathbf{r}_2$ – coordinates of electron 1 on molecule 1 and electron 2 on molecule 2

$r_{21} = |\mathbf{r}_1 - \mathbf{r}_2|$ distance between electrons (**bold** for vectors)

The factor (f^2/n^2) represents (approximately) the local-field effect and dielectric screening in a medium with refractive index n .

9

FRET: electronic interaction

Electronic interaction energy $H_{21(el)}$:

$$H_{21(el)} = \langle \phi_{1a} \phi_{2b} | \tilde{H}_{21} | \phi_{1b} \phi_{2a} \rangle \approx (f^2/n^2) \int \int \phi_{1a}^* \phi_{2b}^* \frac{e^2}{r_{21}} \phi_{1b} \phi_{2a} d\mathbf{r}_1 d\mathbf{r}_2$$

Approximation: express the wavefunctions of ground (a) and excited (b) states as a linear combination of atomic p_z orbitals for *each atom* in molecule 1 and in molecule 2, like this:

$$\psi_{1a} = \sum_s C_s^{1a} \psi_{s,pz}^{1a}$$



This basis is *orthogonal*, and the integral becomes a sum of “transition monopoles”:

$$H_{21(el)} \approx (2f^2/n^2) \sum_s \sum_t C_s^{1a} C_s^{1b} C_t^{2a} C_t^{2b} (e^2/r_{st})$$

Where s and t enumerate atoms in molecule 1 and 2, respectively, and r_{st} is the distance between atoms s and t .

The equation above is simply an electrostatic interaction between the *transition* charge density spread over each molecule. $C_s^{1a} C_s^{1b}$ is the transition charge density for the atoms of molecule 1

10

TDC: Transition Density Cube

Electronic interaction energy $H_{21(el)}$: $H_{21(el)} \approx (f^2/n^2) \int \int \phi_{1a}^* \phi_{2b}^* \frac{e^2}{r_{21}} \phi_{1b} \phi_{2a} d\mathbf{r}_1 d\mathbf{r}_2$

Transition density cube method:

More general: calculate *transition densities* for donor and acceptor based on known wave functions of ground and excited states at every point in the space of donor and acceptor molecules.

Suppose we know the complete wave function of a molecule in ground and excited states:

$$\begin{aligned} \Psi_{1a}(\vec{r}_1, s_1, \vec{r}_2, s_2, \dots) & \leftarrow |\Psi_{1a}|^2 \text{ is the probability to find electron 1 at position } r_1 \\ & \text{in spin state } s_1 \text{ while electron 2 is at } (r_2, s_2) \text{ etc., for all } N \\ \Psi_{1b}(\vec{r}_1, s_1, \vec{r}_2, s_2, \dots) & \text{electrons, in ground state (a)} \end{aligned}$$

Define transition density as probability to find an electron at location $\mathbf{r}$:

$$M_1(\vec{r}) = \int \Psi_{1a}(\vec{r}, s_1, \vec{r}_2, s_2, \dots) \Psi_{1b}^*(\vec{r}, s_1, \vec{r}_2, s_2, \dots) d\vec{r}_2 \cdots d\vec{r}_N ds_1 \cdots ds_N$$

Where integration occurs over all spin coordinates, while integration over $\mathbf{r}$ is for all but one electron, so that $M(\mathbf{r})$ is an average probability of finding one electron at location $\mathbf{r}$.

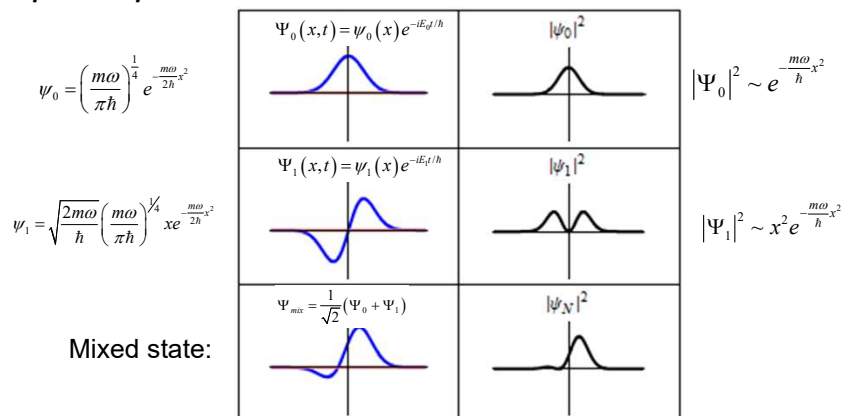
And we have: $H_{21(el)} \approx \frac{f^2}{n^2} e^2 \int \frac{M_1(\vec{r}_1) M_2(\vec{r}_2)}{|\vec{r}_2 - \vec{r}_1|} d\vec{r}_1 d\vec{r}_2$

11

Transition density?

$$M_1(\vec{r}) = \int \Psi_{1a}(\vec{r}, s_1, \vec{r}_2, s_2, \dots) \Psi_{1b}^*(\vec{r}, s_1, \vec{r}_2, s_2, \dots) d\vec{r}_2 \cdots d\vec{r}_N ds_1 \cdots ds_N$$

Simple example: 1D harmonic oscillator



$$|\Psi_{mix}|^2 = \left| \frac{1}{\sqrt{2}} (\Psi_0 + \Psi_1) \right|^2 = \frac{1}{2} |\Psi_0|^2 + \frac{1}{2} |\Psi_1|^2 + |\Psi_0^* \Psi_1|$$

Oscillating term:

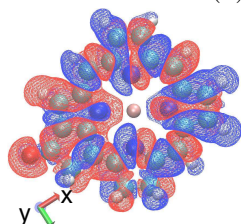
$$|\Psi_0^* \Psi_1| = |\psi_0^* \psi_1| e^{-i(E_1 - E_0)t/\hbar}$$

12

Example: transitional density of Chl a

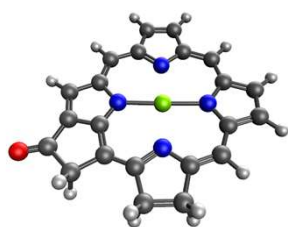
Example: Chl a HOMO→LUMO transition density

$$M_1(\vec{r}) = \int \Psi_{1a}(\vec{r}, s_1, \vec{r}_2, s_2, \dots) \Psi_{1b}^*(\vec{r}, s_1, \vec{r}_2, s_2, \dots) d\vec{r}_2 \cdots d\vec{r}_N ds_1 \cdots ds_N$$



Q: What is the direction of TDM, the transition dipole moment?

1.5706 X 0.6275 Y -0.0007 Z



Calculated by prof. Slipchenko using the simplified Chl a structure from Q-Chem homework, VMD for visualizing

13

TDC: electronic interaction

Electronic interaction energy $H_{21(el)}$: $H_{21(el)} \approx (f^2/n^2) \int \int \phi_{1a}^* \phi_{2b}^* \frac{e^2}{r_{21}} \phi_{1b} \phi_{2a} d\vec{r}_1 d\vec{r}_2$

Transition density cube method:

$$H_{21(el)} \approx \frac{f^2}{n^2} e^2 \int \frac{M_1(\vec{r}_1) M_2(\vec{r}_2)}{|\vec{r}_2 - \vec{r}_1|} d\vec{r}_1 d\vec{r}_2$$

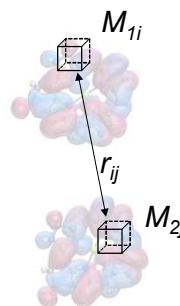
This is the Coulomb interaction term for the *transition density*. Can compute numerically:

- Divide space into small cubes
- Calculate integrals for each cube for each molecule:

$$M_{li} = \int_{cube_i} \Psi_{1a}(\vec{r}, s_1, \vec{r}_2, s_2, \dots) \Psi_{1b}^*(\vec{r}, s_1, \vec{r}_2, s_2, \dots) d\vec{r}_2 \cdots d\vec{r}_N ds_1 \cdots ds_N$$

- Sum up contributions from all "cubes"

$$H_{21(el)} \approx \frac{f^2}{n^2} e^2 \sum_{i,j} \frac{M_{li} M_{2j}}{r_{ij}}$$



See more:

Krueger et al., *J. Phys. Chem. B* **1998**, 102, 5378-5386

Plasser et al, *J.Chem. Phys.* 2014, 141 (<https://doi.org/10.1063/1.4885819>)

14

TDC: electronic interaction

Electronic interaction energy $H_{21(el)}$: $H_{21(el)} \approx (f^2/n^2) \int \int \phi_{1a}^* \phi_{2b}^* \frac{e^2}{r_{21}} \phi_{1b} \phi_{2a} dr_1 dr_2$

Transition density cube method:

$$H_{21(el)} \approx \frac{f^2}{n^2} e^2 \sum_{i,j} \frac{M_{1i} M_{2j}}{r_{ij}}$$

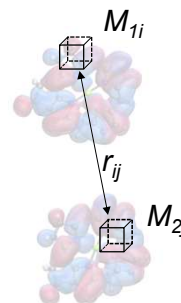
The Coulomb interaction is obvious in this equation

Q: What if each molecule is represented by just one large cube?

- point-monopoles.

We can represent charges in a molecule as a sum of multiple terms – multipole expansion:

monopole + dipole + quadrupole + octupole +

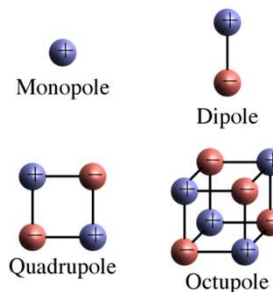
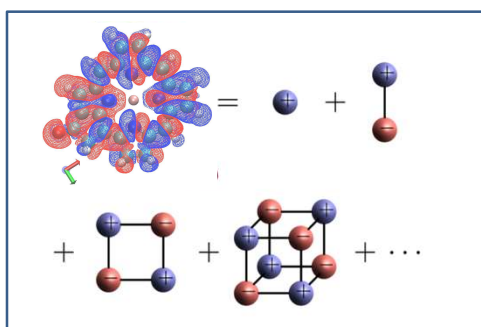


15

Multipole expansion

Multipole expansion:

monopole + dipole + quadrupole + octupole + ...



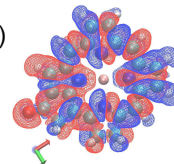
Q: which terms would interact with EM light wave?

16

Transition dipole moment

Transition dipole moment:

The transition dipole moment (dipole term in multipole expansion) can be calculated as before (see Lec01-05, transition dipole moment):



$$\vec{\mu} = e \langle \Psi_a | \vec{r} | \Psi_b \rangle = e \int M(\vec{r}) \vec{r} d\vec{r}$$

$$M(\vec{r}) = \int \Psi_a(\vec{r}, s_1, \vec{r}_2, s_2, \dots) \Psi_b^*(\vec{r}, s_1, \vec{r}_2, s_2, \dots) d\vec{r}_2 \cdots d\vec{r}_N ds_1 \cdots ds_N$$

Transition dipole is prevailing term in multipole expansion for molecules that interact strongly with light.

Note: higher-order contributions can also be calculated from $M(\vec{r})$.

Monopole term is scalar, dipole is a vector, quadrupole is a second-order symmetric tensor (3x3 matrix)

$$Q_{ij} = e \int_V M(\vec{r}) (3x_i x_j - r^2 \delta_{ij}) \vec{r} d\vec{r}$$

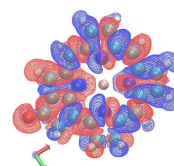
where $x_1=x$, $x_2=y$, $x_3=z$ coordinates

17

Point-dipole approximation

Point-dipole approximation:

Assuming that molecules are far we can replace molecules with "transition" point dipoles.



$$H_{21(el)} \approx \frac{f^2}{n^2} e^2 \int \frac{M_1(\vec{r}_1) M_2(\vec{r}_2)}{|\vec{r}_2 - \vec{r}_1|} d\vec{r}_1 d\vec{r}_2$$

$$H_{21(el)} \approx \frac{f^2}{n^2} e^2 \frac{1}{r_{12}^3} \int M_1(\vec{r}_1) M_2(\vec{r}_2) d\vec{r}_1 d\vec{r}_2$$

Note: this is the same as the electrostatic interaction between charge distributions. Can use classical EM to calculate.

18

Dipole-dipole interaction

Dipole-dipole interaction energy, classical:

Dipole μ_2 in electric field E of dipole μ_1 : $U = -\vec{\mu}_2 \cdot \vec{E}_1$ $\vec{E}_1 = -\vec{\nabla} V_1$

Electric potential at any point for any number of charges q_i distance r_i from that point:

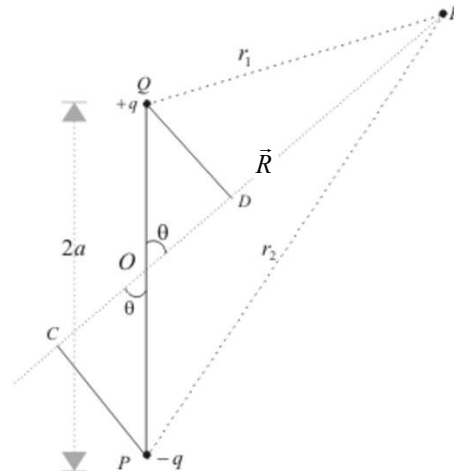
$$V = \frac{q}{r_1} - \frac{q}{r_2}$$

$$V = \frac{q}{R - a \cos \theta} - \frac{q}{R + a \cos \theta}$$

$$V = \frac{2aq \cos \theta}{R^2 - a^2 \cos^2 \theta}$$

But: $a \ll r$, $\mu = 2aq$

$$V = \frac{\mu \cos \theta}{R^2} \equiv \frac{\vec{\mu} \cdot \vec{R}}{R^3}$$



19

Dipole-dipole interaction

Dipole-dipole interaction energy:

Dipole μ_2 in electric field E_1 of dipole μ_1 : $U = -\vec{\mu}_2 \cdot \vec{E}_1$ $\vec{E}_1 = -\vec{\nabla} V_1$

The potential of charge 1 at location of 2: $V_1 = \frac{\vec{\mu}_1 \cdot \vec{R}_{12}}{R_{12}^3}$

Electric field:

$$\vec{E}_1 = -\vec{\nabla} V_1 = -\vec{\nabla} \left(\frac{\vec{\mu}_1 \cdot \vec{R}_{12}}{R_{12}^3} \right) = - \left[\frac{1}{R_{12}^3} \vec{\nabla} (\vec{\mu}_1 \cdot \vec{R}_{12}) + (\vec{\mu}_1 \cdot \vec{R}_{12}) \vec{\nabla} \left(\frac{1}{R_{12}^3} \right) \right]$$

$$\vec{E}_1 = -\frac{\vec{\mu}_1}{R_{12}^3} + 3(\vec{\mu}_1 \cdot \vec{R}_{12}) \frac{\vec{R}_{12}}{R_{12}^5}$$

Dipole-dipole interaction energy:

$$U = -\vec{\mu}_2 \cdot \vec{E}_1 = \frac{\vec{\mu}_2 \cdot \vec{\mu}_1}{R_{12}^3} - \frac{3(\vec{\mu}_1 \cdot \vec{R}_{12})(\vec{\mu}_2 \cdot \vec{R}_{12})}{R_{12}^5}$$

How does it depend on the distance R_{12} ?

20

Dipole-dipole interaction

Dipole-dipole interaction energy:

$$U = \frac{\vec{\mu}_2 \cdot \vec{\mu}_1}{R_{12}^3} - \frac{3(\vec{\mu}_1 \cdot \vec{R}_{12})(\vec{\mu}_2 \cdot \vec{R}_{12})}{R_{12}^5} = \frac{\vec{\mu}_2 \cdot \vec{\mu} - 3(\vec{\mu}_1 \cdot \hat{R}_{12})(\vec{\mu}_2 \cdot \hat{R}_{12})}{R_{12}^3}$$

Unit vector
↓

Corrections: the interaction was derived in a vacuum

- Electric fields are screened by a medium. Transition dipole moment is fluctuating at very high energy of transition: use refraction index, $1/n^2$ is the high-frequency dielectric constant
- Dipoles occupy space in that medium – a cavity. The spherical cavity increases the electric field of the dipole 2 in the surrounding medium by $f \sim 3n^2/(2n^2+1)$, at the same time, and the external field in the cavity of pigment 1 is also increased by the same factor.
Note: f depends on the microscopic properties of the cavities and is generally not easy to find

$n \sim 1.5$ for proteins
 $f \sim$ single digits

Finally, dipole-dipole interaction energy ($H_{21(el)}$) is:

$$H_{21(el)} = \frac{f^2}{n^2} \left[\frac{\vec{\mu}_2 \cdot \vec{\mu}_1}{R_{12}^3} - \frac{3(\vec{\mu}_1 \cdot \vec{R}_{12})(\vec{\mu}_2 \cdot \vec{R}_{12})}{R_{12}^5} \right]$$

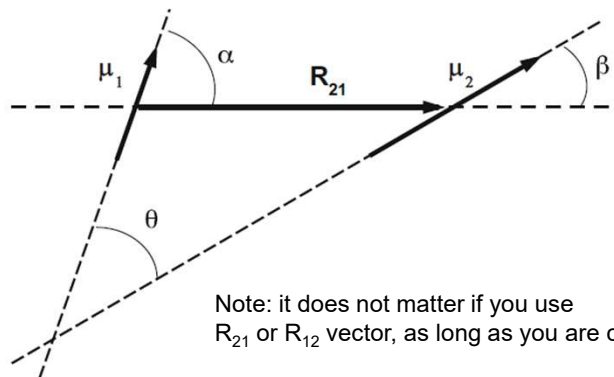
21

Dipole-dipole interaction

Dipole-dipole interaction energy:

$$H_{21(el)} \approx V_{12} = \frac{f^2}{n^2} \left[\frac{\vec{\mu}_2 \cdot \vec{\mu}_1}{R_{21}^3} - \frac{3(\vec{\mu}_1 \cdot \vec{R}_{21})(\vec{\mu}_2 \cdot \vec{R}_{21})}{R_{21}^5} \right]$$

Dipole-dipole interaction energy is usually denoted as V .



Note: it does not matter if you use R_{21} or R_{12} vector, as long as you are consistent

22

Dipole-dipole interaction

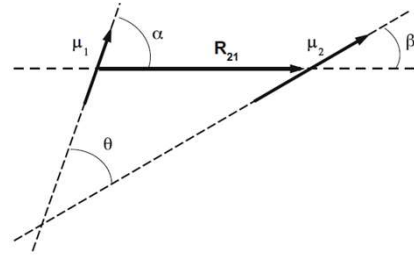
Dipole-dipole interaction energy:

$$V_{12} = \frac{f^2}{n^2} \left[\frac{\vec{\mu}_2 \cdot \vec{\mu}_1}{R_{21}^3} - \frac{3(\vec{\mu}_1 \cdot \vec{R}_{21})(\vec{\mu}_2 \cdot \vec{R}_{21})}{R_{21}^5} \right]$$

Can rewrite in terms of magnitudes:

$$V_{12} = \frac{f^2}{n^2} \left[\frac{\mu_2 \mu_1 \cos \theta}{R_{21}^3} - \frac{3\mu_2 \mu_1 \cos \alpha \cos \beta}{R_{21}^3} \right]$$

$$V_{12} = \frac{f^2}{n^2} \left[\frac{\mu_2 \mu_1 (\cos \theta - 3 \cos \alpha \cos \beta)}{R_{21}^3} \right]$$



Dipole-dipole interaction, point-dipole approximation

$$V_{12} = \frac{f^2}{n^2} \frac{\mu_2 \mu_1 \kappa}{R_{21}^3}$$

$\kappa \equiv \cos \theta - 3 \cos \alpha \cos \beta \leftarrow \text{"orientation factor"}$

23

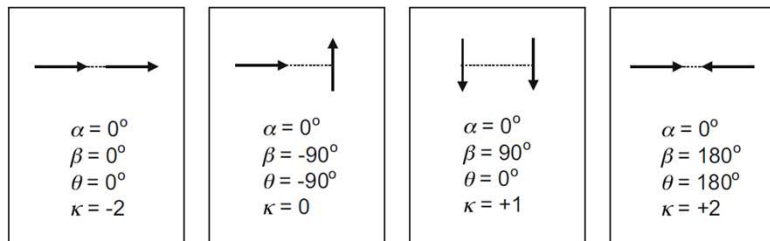
Dipole-dipole interaction

Dipole-dipole interaction energy:

$$V_{12} = \frac{f^2}{n^2} \frac{\mu_1 \mu_2 \kappa}{R_{21}^3}$$

$\kappa \equiv \cos \theta - 3 \cos \alpha \cos \beta \leftarrow \text{"orientation factor"}$

Orientation factor can vary -2...+2



24

Point-dipole: energy transfer rate

Point-dipole energy transfer rate:

$$V_{12} = \frac{f^2}{n^2} \frac{\mu_1 \mu_2 \kappa}{R_{21}^3}$$

$$\kappa \equiv \cos \theta - 3 \cos \alpha \cos \beta$$

$$k_{rt} = \frac{2\pi}{\hbar} |H_{21(el)}|^2 \left\{ \sum_n \sum_m \frac{\exp(-E_{n(1)}/k_B T)}{Z_1} |\langle \chi_{1m} | \chi_{1n} \rangle|^2 \right. \\ \left. \times \sum_u \sum_w \frac{\exp(-E_{u(2)}/k_B T)}{Z_2} |\langle \chi_{2w} | \chi_{2u} \rangle|^2 \right\} \delta(\Delta E_1 - \Delta E_2)$$

The energy transfer rate $\sim V_{12}^2$.

If molecule tumbles randomly need to average over all possible κ^2 : $\langle \kappa^2 \rangle = 2/3$

Energy transfer rate is $\sim 1/R^6$ (in point-dipole approximation)

25

Point-dipole: energy transfer rate

Point-dipole energy transfer rate:

$$V_{12} = \frac{f^2}{n^2} \frac{\mu_1 \mu_2 \kappa}{R_{21}^3}$$

$$\kappa \equiv \cos \theta - 3 \cos \alpha \cos \beta$$

Transition dipole strength is often used instead of transition dipole:

$$D \equiv |\vec{\mu}|^2$$

Absorption is proportional to D; D is usually expressed in Debye² units.

For two identical molecules $\mu_1 = \mu_2 = \mu$:

$$V_{12} = \frac{f^2}{n^2} \frac{\mu^2 \kappa}{R_{21}^3} = \frac{f^2}{n^2} \frac{D \kappa}{R_{21}^3}$$

For D in Debye² and R_{12} in Å:

$$V_{12} = 5.03 \cdot 10^3 \frac{f^2}{n^2} \frac{D \kappa}{R_{21}^3} \quad (V_{12} \text{ in cm}^{-1})$$

26